

Plan for 2025 tracking efficiencies

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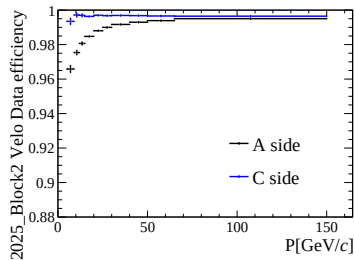
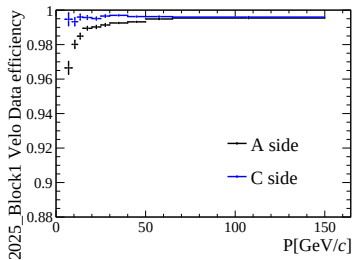
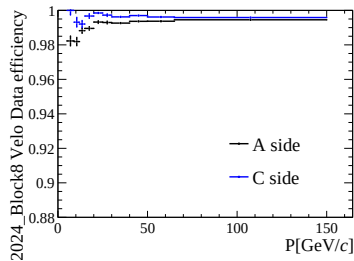
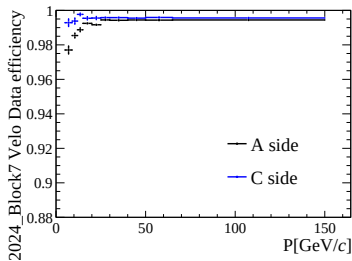


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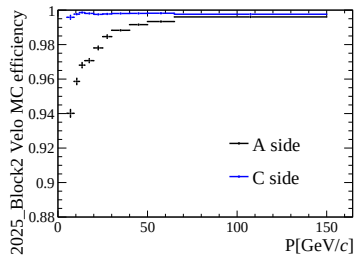
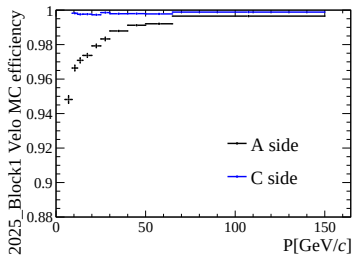
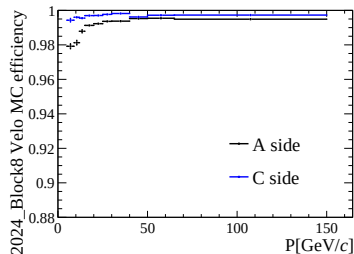
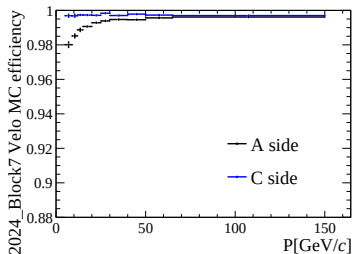
Tracking efficiencies presented at A&S week:

- 2024 tracking efficiencies are available to analysts [here](#)
 - Version 0: Smaller MC samples and with only the main systematic
 - Next release: Updated 2D binning scheme, larger MC samples, multiplicity reweighting
 - Feedback at A&S week: Need finer binning at low p/η due to rapidly changing efficiencies, but this is not really feasible so we need to check if the systematic covers variations within each bin
- 2025 efficiencies are currently not released
 - Velo efficiencies had a large discrepancy between data and MC
 - Unusual behaviour: MC significantly underestimates the tracking efficiency
 - Feedback at A&S week: Check left-right asymmetries and check if M21/M23 are turned off in MC

Velo left-right asymmetry



Velo left-right asymmetry



Tracking efficiencies presented at A&S week:

- M21 and M23 sit at $z = 6.875$ cm and 9.375 cm, respectively
- Each module has 4 sensors
- M23 in data taking:
 - 2024: All sensors excluded
 - 2025 pre-TS: All sensors working
 - 2025 post-TS: 2 sensors working
- M23 in MC:
 - 2024: All sensors excluded
 - 2025 pre-TS: All sensors excluded
 - 2025 post-TS: 2 sensors working
- Furthermore, 2 sensors in M21 were not working in 2025 and also turned off in MC
 - This module is directly in front of M23, so could have significant impact on Velo tracking efficiencies!

What now?

We know 2025 pre-TS MC is wrong, what now...?

- Simulation group knows and MR with fix is ready: [!306](#)
- ... but this is low priority and will not be fixed until Sim10i
 - We're currently on Sim10g!
- Can we ask for this issue to be prioritised?
- Do we need to bin in ϕ ? Possible, but probably requires some work with TrackCalib2
- From my side, apart from this data/MC discrepancy, I am ready to release tracking efficiency correction tables for 2025 pre-TS
- For 2025 post-TS data, I have requested MC [here](#)
 - Current samples are very small (100k), but I've submitted an AP to check left-right asymmetry
 - Should in principle have a better data/MC agreement

Open questions:

- ① Is it necessary to bin in ϕ ?
 - Perhaps with the larger MC samples, we can have two ϕ bins for the charge integrated results, and for charge separated results we have no ϕ binning
- ② RTA performance paper?
 - What figure is needed?
- ③ Internal note on tracking efficiencies?
 - Started by Rowina, but a lot more work is needed to finalise